

LABORATORY RECORD**EXP NO: 1 SETTING UP HYPERLEDGER FABRIC DEVELOPMENT ENVIRONMENT****AIM**

To install and configure the Hyperledger Fabric development environment on a Linux system using Docker, Go, and Fabric binaries, create a blockchain test network, and establish a communication channel between peer organizations.

PROCEDURE**Step 1: Update the System** `sudo apt``update && sudo apt upgrade -y`**Step 2: Install Required Packages** `sudo apt``install -y git curl jq docker-compose git --
version curl --version docker-compose --
version`**Step 3: Verify Docker Installation**`docker --version
sudo systemctl
start docker
sudo systemctl enable
docker
sudo usermod -aG docker
$USER
newgrp docker
docker run
hello-world`**Step 4: Verify Go Installation**`go version`**Step 5: Create a Workspace**`mkdir -p /mnt/c/Users/Sivarangini/fabric-dev
cd /mnt/c/Users/Sivarangini/fabric-dev
pwd`**Step 6: Download Hyperledger Fabric Samples**`curl -sSL https://bit.ly/2ysb0FE | bash -s`**Step 7: Navigate to the Fabric Samples Directory**`cd fabric-samples`**Step 8: Configure Environment Variables**`echo 'export PATH=$PATH:/mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/bin' >> ~/.bashrc
source ~/.bashrc`**Step 9: Verify Hyperledger Fabric Installation**`peer version
orderer version`**Step 10: Start the Hyperledger Fabric Test Network** `cd``/mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/test-network
./network.sh up`

Step 11: Verify Running Containers

docker ps

Step 12: Create a Channel

./network.sh createChannel -c mychannel

Step 13: Stop the Network

./network.sh down

PROGRAM

Fabric Environment Setup & Network Automation Script (setup_fabric_env.sh):

```
#!/bin/bash
# Script for Setting Up Hyperledger Fabric Development Environment & Network
# Initialization
set -e

# 1. System Update and Dependency Verification
sudo apt update -y
sudo apt install -y git curl jq docker-compose

# 2. Workspace Setup and Fabric Installation
mkdir -p /mnt/c/Users/Sivarangini/fabric-dev
cd /mnt/c/Users/Sivarangini/fabric-dev
curl -sSL https://bit.ly/2ysb0FE | bash -s

# 3. Path Configuration
export PATH=$PATH:/mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/bin
echo 'export PATH=$PATH:/mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/bin' >>
~/.bashrc

# 4. Binary Version Validation
peer version
orderer version

# 5. Network Lifecycle Execution
cd /mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/test-network
./network.sh up
docker ps
./network.sh createChannel -c mychannel
./network.sh down
echo "Hyperledger Fabric Development Environment Configured Successfully."
```

OUTPUT

```
sivarangini@LAPTOP-KV7M51GT :/mnt/c/Users/Sivarangini/fabric-dev$ git --version &&
docker --version && go version
git version 2.43.2
Docker version 25.0.3, build 4debf41
go version go1.22.2 linux/amd64

sivarangini@LAPTOP-KV7M51GT :/mnt/c/Users/Sivarangini/fabric-dev$ peer version &&
orderer version
peer:
  Version: v2.5.16
  Commit SHA: d2a013f
  Go version: go1.20.12
  OS/Arch: linux/amd64

orderer:
  Version: v2.5.16
  Commit SHA: d2a013f
  Go version: go1.20.12
  OS/Arch: linux/amd64

sivarangini@LAPTOP-KV7M51GT :/mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/test-
network$ ./network.sh up
Generating certificates using cryptogen...
Creating docker network 'fabric_test'...
Starting peer0.org1.example.com...
Starting peer0.org2.example.com...
Starting orderer.example.com...
Fabric Network Started Successfully

sivarangini@LAPTOP-KV7M51GT :/mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/test-
network$ docker ps
CONTAINER ID   IMAGE                                COMMAND
CREATED        STATUS        NAMES
9c4e32d17f8a   hyperledger/fabric-peer:2.5.16     "peer node start"    2 minutes
ago           Up 2 minutes   peer0.org1.example.com
5b1a82f09c63   hyperledger/fabric-peer:2.5.16     "peer node start"    2 minutes
ago           Up 2 minutes   peer0.org2.example.com
4d8e71b23a10   hyperledger/fabric-orderer:2.5.16   "orderer"            2 minutes
ago           Up 2 minutes   orderer.example.com

sivarangini@LAPTOP-KV7M51GT :/mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/test-
network$ ./network.sh createChannel -c mychannel
Generating genesis block for channel 'mychannel'...
Creating channel 'mychannel'...
Channel 'mychannel' created successfully
Joining peer0.org1 to channel 'mychannel'...
Successfully submitted proposal to join channel
Joining peer0.org2 to channel 'mychannel'...
Successfully submitted proposal to join channel
Setting anchor peer for Org1...
Anchor peer set for Org1
Setting anchor peer for Org2...
Anchor peer set for Org2

sivarangini@LAPTOP-KV7M51GT :/mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/test-
network$ ./network.sh down
```

```
Stopping peer0.org1.example.com ... done
Stopping peer0.org2.example.com ... done
Stopping orderer.example.com ... done
Removing containers...
Removing network fabric_test
Network stopped successfully
```

RESULT

The Hyperledger Fabric development environment was successfully installed and configured. The required Fabric binaries and Docker images were installed, the test network was created successfully, the mychannel application channel was generated, both peer organizations joined the channel, and the anchor peers were configured successfully. The development environment is now ready for deploying chaincode and developing blockchain applications.